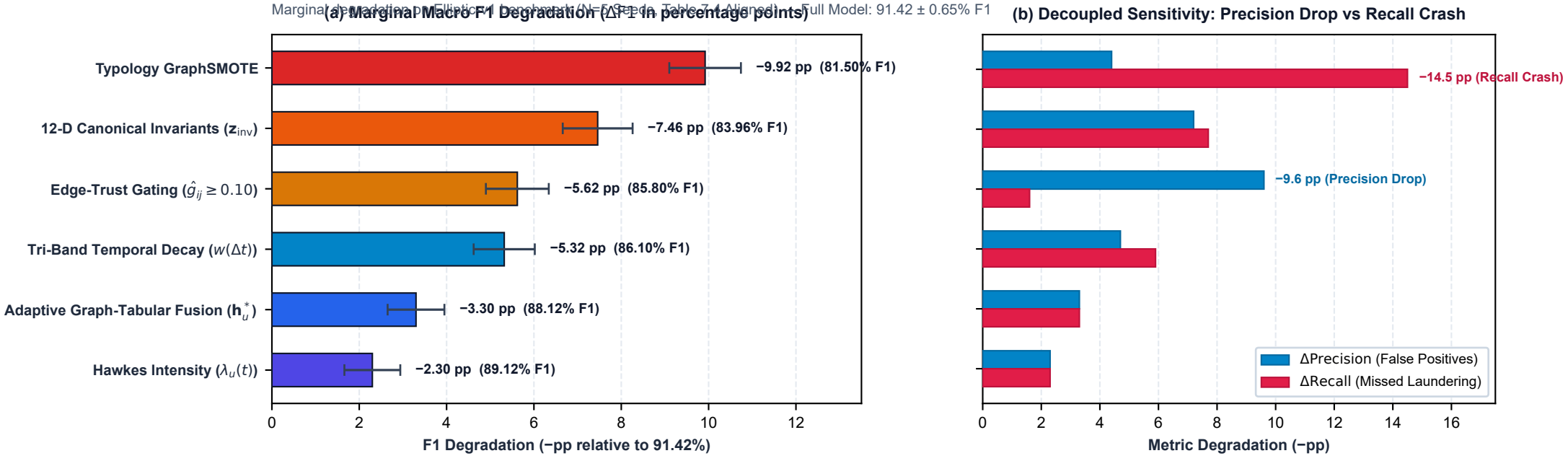
Leave-One-Out Component Removal Waterfall & Asymmetry Analysis

Marginal degradation on Elliptic++ benchmark ($N = 5$ seeds, Table 7.1 Aligned) — Full Model: 91.42 ± 0.65% F1



Architectural Component	Marginal Impact	Theoretical & Operational Failure Mode	Statutory Risk
Typology GraphSMOTE	**ΔF1: −9.92 pp** **ΔRec: −14.5 pp**	Severe minority class collapse under 0.05% skew; failure to learn motifs	High FN Regulatory Fines
12-D Canonical Invariants	**ΔF1: −7.46 pp** **Symmetric drop**	Violation of flow conservation & capital velocity kinetics across hops	Cross-Rail Non-Transfer
Edge-Trust Gating	**ΔF1: −5.62 pp** **ΔPrec: −9.6 pp**	Spurious camouflage chaff overwhelms message passing (smurfing noise)	Severe Alert Fatigue
Tri-Band Decay + Hawkes	**ΔF1: −7.62 pp** **Time sensitivity**	Inability to decouple microsecond transaction bursts from 90-day dormancy	Evasion via Temporal Gap